1. Wall conduction and thermal transmittance

For steady conduction through a uniform slab, heat-transfer power increases with area, temperature difference and conductivity, and decreases with thickness. Thermal transmittance combines material and thickness effects. A smaller value generally indicates better insulation. The simple single-layer model neglects surface resistances. Transmittances of successive layers cannot simply be added.

$$\frac{Q}{t} = \frac{\kappa A(T_H - T_C)}{d}$$

$$U = \frac{\kappa}{d}$$

Steady conduction through a uniform slab. κ: thermal conductivity (W m⁻¹ K⁻¹); A: m²; d: thickness (m); temperature difference: K or °C; Q/t: W.

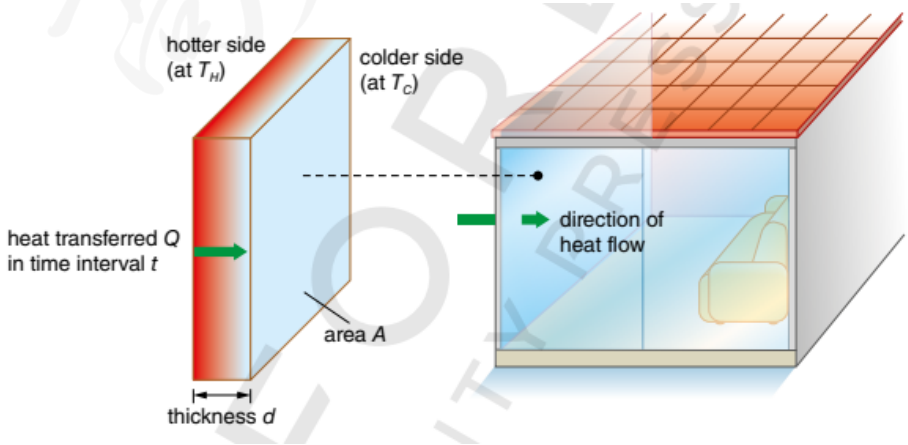
Single homogeneous layer, neglecting surface resistances. U: thermal transmittance (W m⁻² K⁻¹); κ: W m⁻¹ K⁻¹; d: thickness (m).

2. Overall thermal transfer value

Overall thermal transfer value is heat gained through the considered building envelope, from conduction and solar radiation, divided by its total area. It compares average heat gain and is measured in W m⁻². Thermal transmittance also divides by temperature difference, so the definitions differ. Use the stated definitions when combining wall, shading and glazing gains.

$$\text{OTTV} = \frac{Q_C/t + Q_S/t}{A_T}$$

Overall thermal transfer value (W m⁻²); Q_C/t: conductive heat gain (W); Q_S/t: solar heat gain (W); A_T: total area of the building envelope being considered (m²).



Heat flows from the hotter side of the wall to the colder side. Thickness is measured along heat flow.

3. Low-emissivity glass and solar-control films

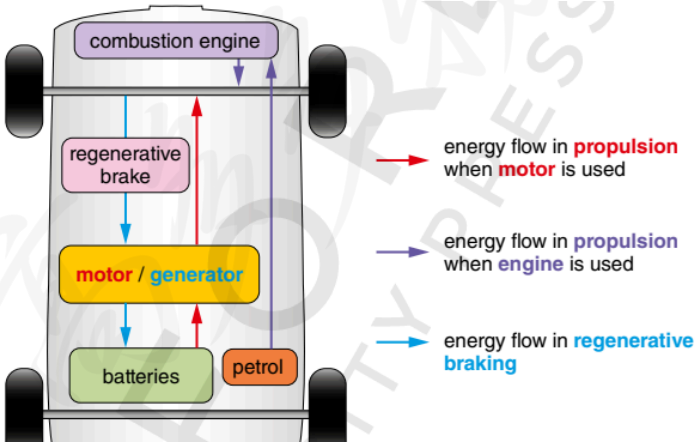
The textbook comparison describes glazing and films that preferentially reflect infrared and ultraviolet while transmitting much visible light. Visible transmission provides natural lighting, while reducing unwanted thermal radiation lowers heat gain.

4. Regenerative braking

During regenerative braking, the drive system acts as a generator, converting part of the vehicle's kinetic energy into electrical energy stored chemically in the battery. This reduces energy directly dissipated as heat by friction brakes, but conversion and storage losses prevent complete recovery. Energy flow is opposite to battery-powered acceleration.

5. Engine and motor roles in hybrid vehicles

A hybrid can be driven by its engine, motor or both. The textbook example uses both during acceleration or uphill travel, and motor assistance can permit a smaller engine. This division of propulsion differs from regenerative braking, which recovers energy during deceleration. Match the mechanism to the operating stage.



Energy flows between the engine, electric motor and battery in a hybrid vehicle.